🚀 DevOps Roadmap 2026: Zero to ₹20 LPA (Step-by-Step)

🎯 Goal of this Roadmap

✅ Build job-ready DevOps skills from scratch
✅ Create a strong **portfolio (projects + GitHub)**
✅ Prepare for interviews with real-world tools & workflows
✅ Become confident in **cloud, CI/CD, containers, Kubernetes, IaC, observability, security, and SRE**

🧱 Phase 1: Foundations (Linux & Git)

🐧 1. Linux

✅ What you must learn

- 🧭 **Basics:** Commands, File System, Permissions, Processes
- 👥 **Users & Groups:** useradd/usermod, groups, sudoers
- 🔐 **Remote Access:** SSH keys, hardening, common issues
- 🌐 **Networking:** IP, ports, `netstat`, `ss`, `ping`, `curl`, `traceroute`
- 📦 **Package Management:** `apt/yum`, tar, gzip, wget
- 🪄 **Scripting:** Bash scripts, loops, variables, functions, cron jobs
- 📊 **System Monitoring:** `top`, `htop`, `df`, `du`, `free`, `journalctl`

🏁 Goal (Outcome)

✅ Be comfortable **managing Linux servers** and **automating tasks** like a real DevOps engineer.

🌿 2. Git

✅ What you must learn

- 🗂️ **Version control basics:** clone, commit, push, pull, branch
- 🌳 **Branching:** feature branches, GitFlow basics

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- 🔀 **Merge & Rebase:** when to use what
- ⚔️ **Conflict resolution:** resolving merge conflicts
- 🔍 **Pull Requests & Code Reviews:** approvals, comments, changes
- 🪝 **Git hooks & automation:** pre-commit checks, linting, formatting

🏁 Goal (Outcome)

✅ Collaborate efficiently with teams and support **CI/CD pipelines** with clean Git practices.

☁️ Phase 2: Cloud Fundamentals (AWS)

🟧 3. AWS Core Services

✅ What you must learn

- 🖥️ **Compute:** EC2, Lambda, Auto Scaling
- 🌐 **Networking:** VPC, Subnets, Route53, Security Groups, NACL
- 🛡️ **Identity & Access:** IAM Users, Roles, Policies, MFA
- 🗄️ **Storage:** S3, EBS, EFS, Glacier
- 📈 **Monitoring:** CloudWatch, CloudTrail, Logs, Alerts
- 📦 **Containers:** EKS, ECS
- 🧰 **Essentials:** Session Manager (SSM), Load Balancers (ALB/NLB)

🏁 Goal (Outcome)

✅ Deploy, secure, and monitor applications on AWS like production environments.

📦 Phase 3: Containerisation & Orchestration

🐋 4. Docker

✅ What you must learn

- ⚙️ Docker Engine, Images, Containers, Volumes, Networks
- 📄 Dockerfile best practices

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- 🧱 Multi-stage builds
- 🧩 Docker Compose for multi-container apps
- 👷 Docker Swarm basics *(optional)*

🏁 Goal (Outcome)

✅ Package applications into **portable containers** ready for deployment.

☸️ 5. Kubernetes (Core + EKS/AKS)

✅ What you must learn

- 🧊 **Core Concepts:** Pods, Services, Deployments, ReplicaSets
- 🏷️ **Namespaces, Labels, Selectors**
- 🧩 **ConfigMaps & Secrets**
- 🩺 **Debugging:** logs, exec, describe, events
- 🌐 **Networking:** Ingress, Load Balancing, Service Types
- 💾 **Storage:** PV/PVC, StorageClasses (basics)
- ☁️ **EKS & AKS:** setup, node groups, networking overview
- 📏 **Autoscaling:** HPA, VPA, Cluster Autoscaler

🏁 Goal (Outcome)

✅ Deploy, scale, and manage containerised applications at **production scale**.

📦 6. Helm (Kubernetes Packaging)

✅ What you must learn

- 🧱 Helm chart structure (Chart.yaml, values.yaml, templates/)
- 🧩 Templates & variables (helpers, conditionals, loops)
- 🏷️ Versioning & releases
- ↩️ Rollbacks and upgrades
- 🗂️ Environment-based configs (dev/stage/prod values files)
- ✅ Helm best practices (reusability, naming, defaults)

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🏁 Goal (Outcome)

✅ Package and deploy applications to Kubernetes **repeatably** with versioned releases.

🔄 7. Argo CD (GitOps Continuous Delivery)

✅ What you must learn

- 🌿 GitOps concepts: Git as the source of truth
- 🔄 Sync policies (manual vs auto-sync)
- 🧭 App health, drift detection, self-heal
- 🚀 Progressive delivery basics (rollbacks, version pinning)
- 🧩 Deploy Helm charts using Argo CD
- 🔐 RBAC and access control (team-based deployments)

🏁 Goal (Outcome)

✅ Achieve automated, reliable Kubernetes deployments using **Git-based workflows**.

🏗️ Phase 4: Infrastructure as Code & Configuration Management

🧱 8. Terraform

✅ What you must learn

- 📝 HCL basics: Providers, Resources, Variables, Outputs
- 🧩 Modules & Workspaces
- 🗃️ Remote State & Backends (S3, Terraform Cloud)
- 🔄 Lifecycle management & dependency graphs
- ☁️ Terraform + AWS: VPC, EC2, IAM, S3 automation

🏁 Goal (Outcome)

✅ Provision cloud infrastructure **reproducibly and safely**.

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🛠️ 9. Ansible

✅ What you must learn

- 📋 Inventory, Playbooks, Roles
- 🧩 Modules: file, package, service, template
- 🔁 Variables, Facts, Conditionals, Loops
- 🖥️ Ansible Tower / AWX *(optional)*
- 🔗 Integrate with CI/CD pipelines

🏁 Goal (Outcome)

✅ Automate server configuration and deployments at scale.

🔄 Phase 5: CI/CD & DevOps Automation

🧪 10. GitLab / Jenkins

✅ What you must learn

- 🏗️ GitLab pipelines, runners, jobs, artefacts
- 🧰 Jenkins pipelines (declarative + scripted)
- 🚦 Blue/Green and Canary deployments
- 📦 Nexus / Artifactory basics
- 🔗 Integration with Terraform, Docker, Kubernetes, Helm, Argo CD

🏁 Goal (Outcome)

✅ Build fully automated CI/CD pipelines from commit to production deployment.

📊 Phase 6: Observability & Monitoring

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🟪 11. DataDog

- 📈 Metrics, dashboards, monitors, alerts
- ☁️ AWS + ☸️ Kubernetes integrations
- 🧠 APM basics (traces, latency, error rates)

🟫 12. ELK Stack

- 🧲 Logstash/Beats for log collection
- 🗃️ Elasticsearch for storage and search
- 📊 Kibana dashboards + visualisation

🏁 Goal (Outcome)

✅ Monitor, alert, troubleshoot, and debug systems effectively.

📨 Phase 7: Messaging & Streaming

🟦 13. Kafka / MSK / Confluent

- 🧠 Topics, producers, consumers, partitions
- 🔌 Kafka Connect, Kafka Streams *(intro)*
- ☁️ Managed Kafka: MSK / Confluent
- 💡 Use cases: event-driven systems, logging pipelines, analytics pipelines

🏁 Goal (Outcome)

✅ Understand streaming pipelines and operations from a DevOps point of view.

🔐 Phase 8: Security & Reliability

🔒 14. SSL Certificates

- 🧾 Self-signed vs CA-signed
- 🌐 HTTPS setup in Nginx / Apache
- ☁️ AWS ACM for managed certificates

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🧯 15. SRE

- 🔪 SLIs / SLOs / SLAs
- 🚑 Incident response + runbooks
- 🧨 Chaos engineering *(intro)*
- 🧾 Postmortems + reliability metrics

🛡️ 16. Security Tools

- 🧪 SAST/DAST: SonarQube, OWASP ZAP
- 🐋 Container security: Trivy, Aqua, Anchore
- ☁️ Cloud security best practices (IAM, logging, secrets)

🏁 Goal (Outcome)

✅ Ensure systems are **secure, reliable, resilient**, and production-ready.

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